Experiment no: 1

Familarization with AND, OR and Inverter plan

Objectives:

1) Investigate the relationship between inputs and outputs and AND Gate
2) Examine the input and output of inverter Gate
3) Investigate the relationship between inputs and outputs of OR Gate.

Basic Theory:

AND Gate → A logic circuit whose output is only high when all inputs are high. It takes two input and gives one output.

fig: AND Gate

Truth table of AND Gate

A	B	A * B
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate → A logic circuit whose output is high when any of the inputs are high. It takes two input and gives one output.

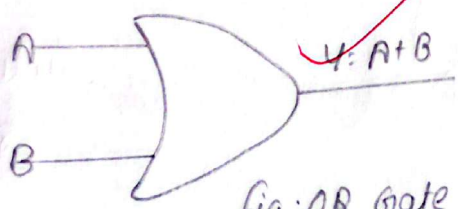

fig: OR Gate

Truth table of OR Gate

A	B	A+B
0	1	1
1	0	1
1	1	1
0	0	0

Inverter gate → A logic circuit whose output is high when input is low and vice versa. It is also known as NOT Gate. It only takes one input and gives one output.

fig: NOT Gate

Truth table of NOT Gate

A	$\bar{A}$
0	1
1	0

Diagrams and Observations

Objective no.1 → Investigate the relationship between inputs and outputs of AND Gate

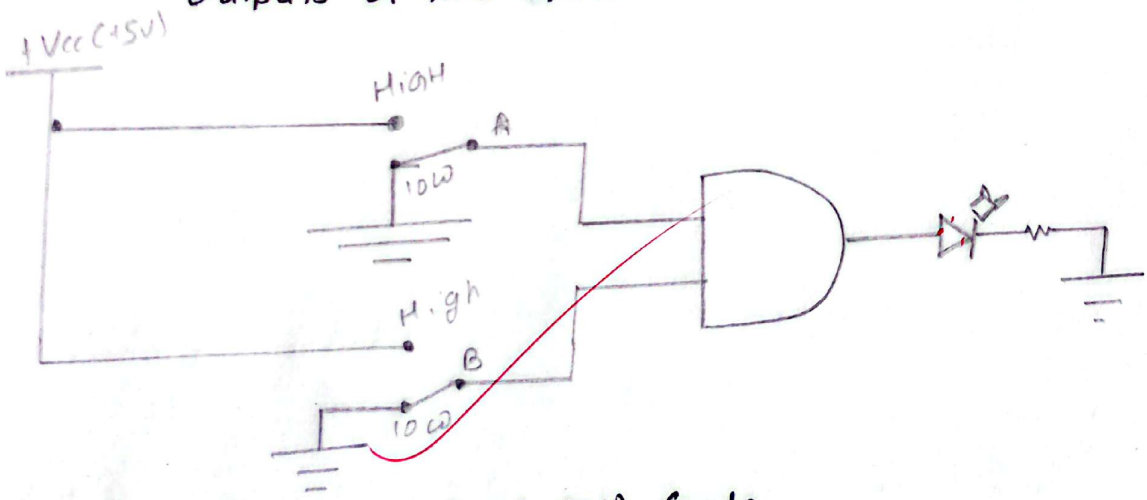

Fig: 2 Input circuit of AND Gate.

Truth Table:

A	B	fe(Expected output)	f(Observed Output)
0	0	0	0
0	1	0	0
1	0	0	0
1	1	1	1

Expected and observed Outputs are Same

Objective no 2:

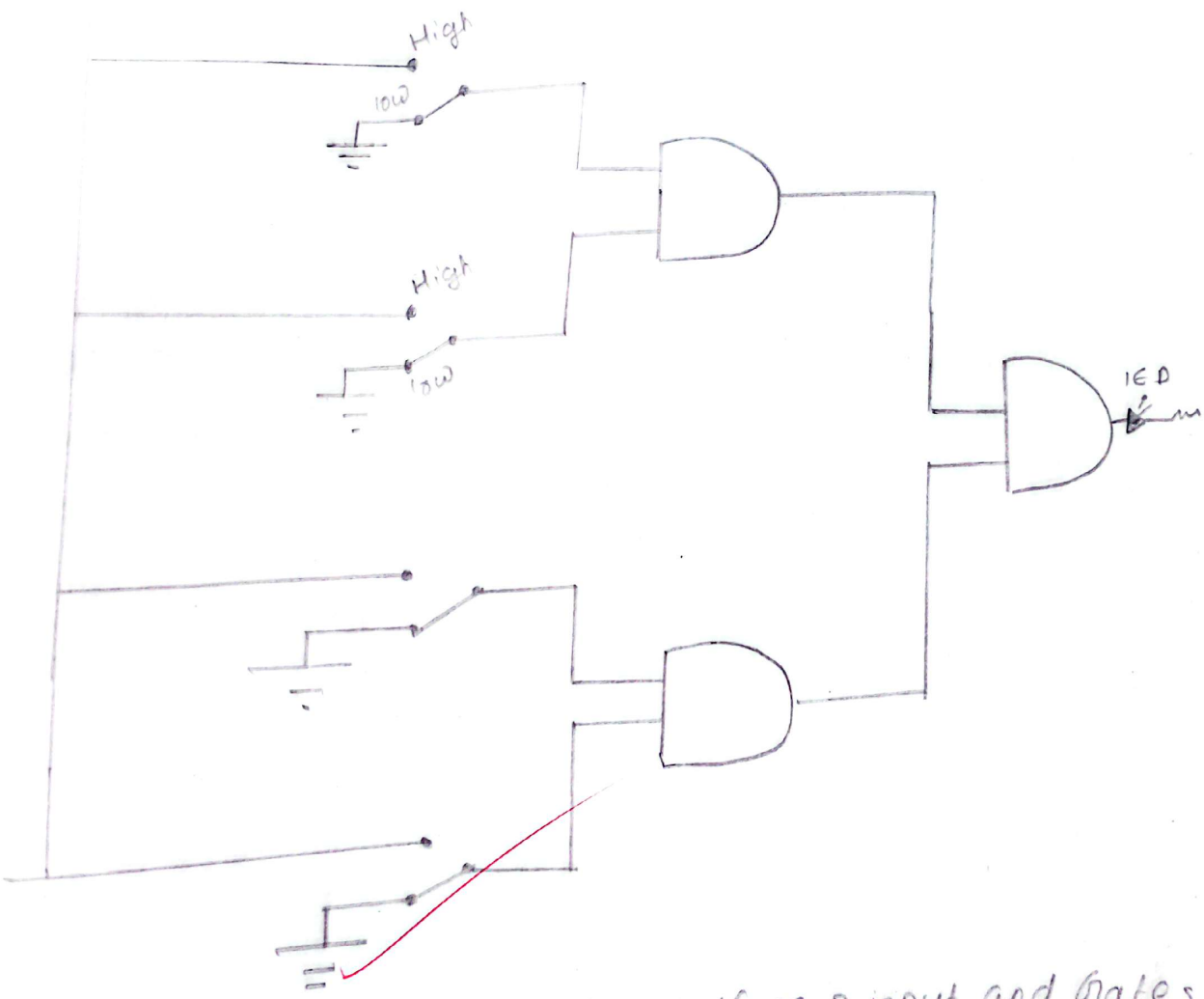

fig: 4 Input AND from 8 three 2 input and Gates circuit.

Truth table

A	B	C	D	A*B	C*D	fe	f
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0
0	0	1	1	0	1	0	0
0	1	0	0	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	0	0	0	0	0
0	1	1	1	0	1	0	0
1	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0
1	0	1	0	0	0	0	0
1	0	1	1	0	1	0	0
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	0
1	1	1	0	1	0	0	0
1	1	1	1	1	1	1	1

Expected output and observed Output are the same.

Objective NO:3 → Investigate the relationship between input & outputs of OR gate.

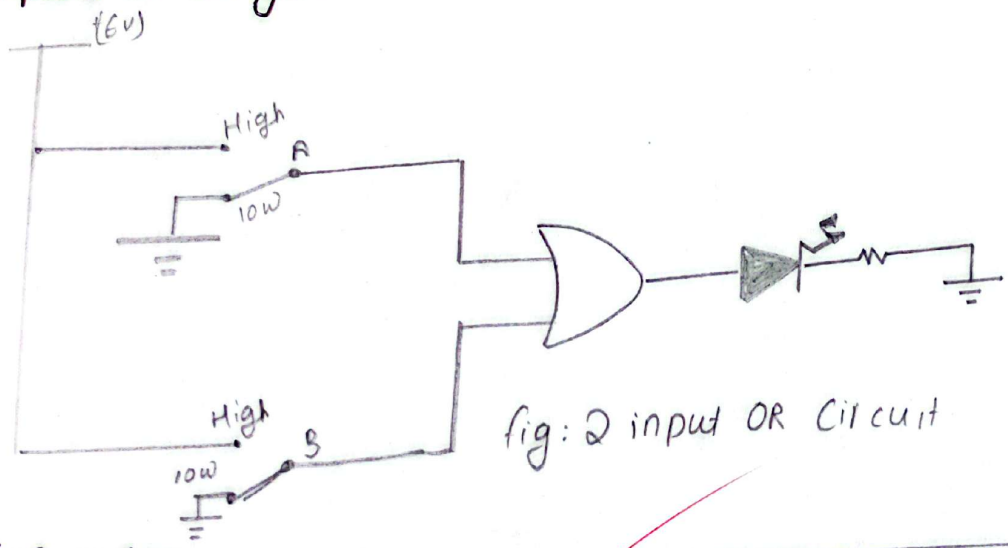

fig: 2 input OR Circuit

Truth table

A	B	fe (Expected Output)	f (Observed output)
0	0	0	0
0	1	1	1
1	0	1	1
1	1	1	1

Objective no 4: Examine the Input and output of inverter Gate

Case 1.

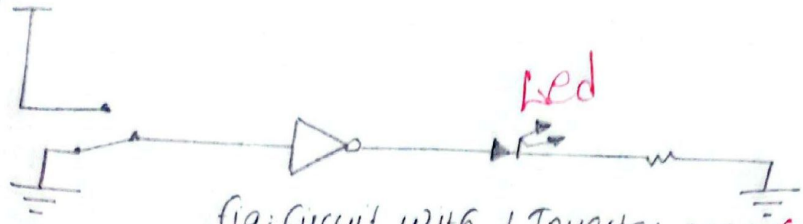

fig: Circuit with 1 Inverter gate GND

Truth table:

A	f
0	1
1	0

Case - 2

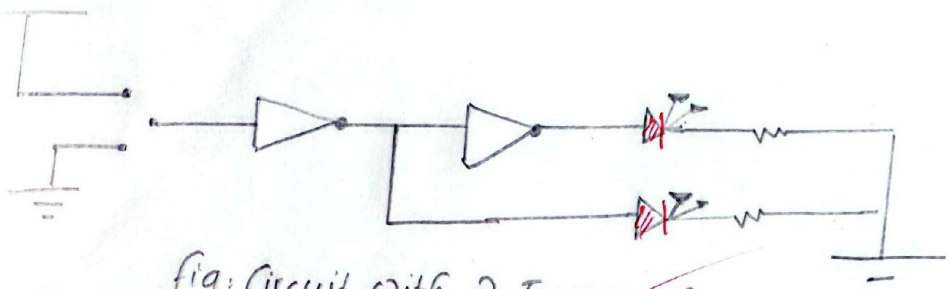

fig: Circuit with 2 Inverter Gates

Truth table

A	~~B~~ f_1	f_2
0	1	0
1	0	1

Case - 3

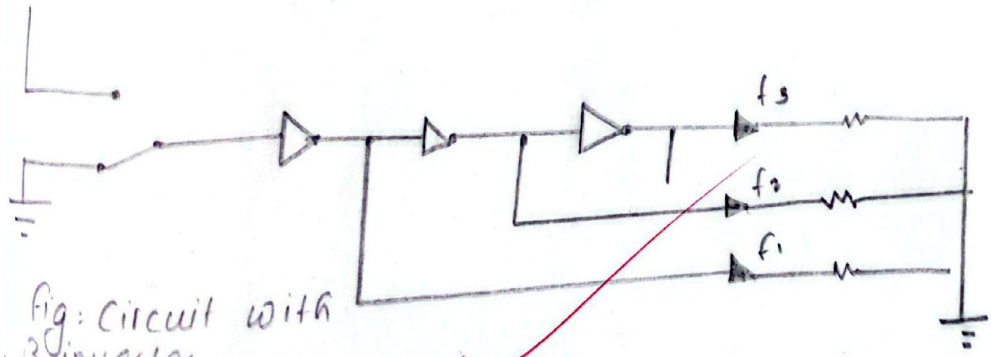

fig: Circuit with 3 inverter

A	f_1	f_2	f_3
0	1	0	1
1	0	1	0

Conclusion

-) In this lab session, we learned about different types of gates and how they work accordingly. Therefore, AND gates only result a high output when all input is high while OR gate result High output when any input is high and Not gate always gives complement of itself as result.

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